Metamorphic Testing of a Housing Trends Application

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System Overview

A multiagent application was built to report twelve-month housing trends for user-selected criteria. It was evaluated using ten metamorphic relations. The completed experiment contains 900 paired comparisons. All 690 pairs with two valid numerical outputs satisfied their declared relationships. The remaining 210 pairs had output-contract failures, which were investigated and reproduced.

The application uses a synthetic Dallas-themed dataset generated locally. Users can choose ZIP codes, property types, bedroom counts, floor-area bounds and price bounds. The result contains monthly observation counts, median asking prices and median prices per square foot, together with percentage changes and optional ZIP comparisons.

Application Design

One Orchestrator receives the application request and returns the final result. It delegates work to Price, Supply, Region, Reviewer and Composer agents through Fast Agent. Region runs when the user requests regional comparison or when a review requires additional diagnosis.

Each role makes a real model call and returns a structured decision. Python validates that decision and executes the permitted operation. Price, Supply and Region request statistical tools. Reviewer decides whether to accept the result, request another scan, accept limited support or fail. Composer selects verified facts and the required caveats for the final explanation.

The controller keeps the dataset and user criteria fixed during an analysis. It checks role permissions, execution order and repair limits. Statistical aggregation uses the completed specialist receipts. An invalid or incomplete execution ends with an explicit error.

User Interface

View	Purpose
Explore	Choose criteria, inspect monthly trends, change display units and export CSV or JSON.
Workflow	Inspect the agent roles, decisions, tool results, feedback and failures.
Experiments	Review the ten MR results and open paired cases and transformation witnesses.
Provenance	Review the data source, generation method and statistical definitions.

The frontend uses React and TypeScript. The API, command-line interface and experiment harness call the same run_analysis service. Live analysis uses a local Qwen model through Ollama. The frozen experiment used Python 3.15.0 and Fast Agent 0.10.45.

Dataset and Output Metrics

A total of 120 observations per ZIP per month were generated for twelve ZIP codes from October 2024 through September 2026. This gives 34,560 observations. The additional history supports comparisons with months before the visible twelve-month window.

A seeded Python generator creates the observations. It assigns a property type, one to five bedrooms and a floor area. It derives prices from floor area and a generated price per square foot, with ZIP and property-type differences, a designed time trend, seasonal variation and random noise. These values describe a software test scenario. They are not measured Dallas market observations.

The generator uses seed 20261016. Two independent regenerations produced identical CSV bytes. The saved manifest records the seed, recipe, row counts, month coverage and file hashes. Each observation has a unique record ID. Listing labels repeat across months, so the counts measure observations rather than unique homes.

Metric	Definition
Monthly count	Number of observations that match all criteria in a visible month.
Median price	Median asking price among the matching observations.
Median price per area	Median of each observation's price divided by its floor area.
Monthly changes	Price and price-per-area month-over-month and year-over-year percentages, plus count month-over-month percentage.
Window summary	Total observations and first-to-last price and price-per-area percentage changes.
Final month summary	The final month's price and price-per-area year-over-year changes.
ZIP comparison	Median price per square foot across each requested ZIP's observations in the visible window, with ordinal ranking.

Prices use integer cents and calculations use exact fractions. Values serialize to six decimal places with half-even rounding. An empty month has count zero and unavailable medians. A percentage is unavailable when its comparison value is missing or zero. First-to-last change covers eleven monthly intervals within the twelve-month window.

For example, prices of $200,000 and $400,000 with areas of 1,000 and 4,000 square feet give individual ratios of $200 and $100 per square foot. Their median is $150. Dividing the median price by the median area would give $120. The application uses the individual-ratio calculation.

Agent Workflow

The initial plan uses the indexed dataset and always activates Price and Supply. It also activates Region when regional comparison is requested. The specialists return

validated tool requests. Python runs the tools on the controller-bound selection and records receipts showing the query, selected records and result identity.

After the active specialists finish, Python checks their receipts and aggregates the statistics. Reviewer receives the actual coverage and integrity signals. A complete result can be accepted. Sparse support or incomplete indexed coverage can trigger a full CSV scan with Region diagnosis. Confirmed sparse support leads to a limited result with a caveat.

The Orchestrator must apply the Reviewer's requested change. It cannot broaden price bounds, add ZIPs or otherwise change the user's query. A later new receipt mismatch permits one rebuilt-index pass. The controller allows at most two refinements and one schema repair for an invalid decision. Persistent integrity problems cause failure.

Observed Feedback Loop

A natural campaign source run under seed 104729 contained 35 visible observations and 63 observations including the available comparison months. Its initial indexed scan produced a sparse-support signal. The following sequence occurred in the recorded run.

Step	Recorded action
1	Orchestrator planned an indexed scan with Price and Supply.
2	Price and Supply completed their tool requests. Python aggregated their results.
3	Reviewer returned REFINE with SPARSE_SLICE.
4	Orchestrator changed to a full CSV scan and activated Region for coverage diagnosis.
5	Fresh Price, Supply and Region executions confirmed the same selection and statistics.
6	Reviewer returned ACCEPT_LIMITED with CONFIRMED_SPARSE. Composer included the applicable caveats and the parent completed the run.

The run made sixteen model decisions. The user criteria and canonical statistics stayed unchanged across the two passes. The full scan confirmed sparse data rather than finding extra observations. The plans and specialist receipts changed because the system performed another scan.

A separate auxiliary test deliberately removed a qualifying observation from the index. Its full scan recovered that observation. This injected control was kept outside the headline experiment. The natural sparse example above demonstrates the feedback loop without an injected fault.

Metamorphic Relations

A metamorphic relation specifies how selected outputs should relate after a controlled change to the input. The applicable inputs are defined and each

transformation is checked for a meaningful effect before the agents run. Narrative wording, run IDs and workflow routes are outside the numerical comparisons.

MR01 Row Permutation

Family: invariance. Every raw row is shuffled while all fields and multiplicities are preserved. The test requires a changed row order, at least two matching observations and a defined visible median. All monthly and summary statistics, including requested ZIP rankings, must match exactly. This checks for order-dependent selection, grouping, medians and indexing.

MR02 Identifier Relabeling

Family: invariance. Record and listing IDs are renamed using one-to-one mappings. Repeated occurrences of the same listing label receive the same new label. Geography and numerical fields remain fixed. At least one selected identifier must change. All housing statistics must match exactly. This checks for accidental filtering, deduplication or numerical interpretation based on identifiers.

MR03 Observation Replication

Family: invariance of distribution. Every observation is copied once. Each copy receives a fresh record ID and retains its listing label. The raw row count and each selected multiplicity double. Monthly counts and the window total must double. Medians, price changes, count percentage changes and ZIP rankings must remain unchanged. This checks for unintended listing-label deduplication and incorrect aggregation of repeated values.

MR04 Display Units

Family: invariance. The data and criteria remain fixed while the display switches from dollars to thousands of dollars and from square feet to square metres. Canonical statistics and ranks must match exactly. Displayed prices divide by 1,000. Displayed prices per square metre use the area conversion as well. The display tolerance is 0.000001 in the displayed unit. This checks unit conversion, early rounding and filtering in display units.

MR10 Excluded ZIP Observations

Family: invariance. Valid observations are appended in an allowed ZIP excluded by the query. The new records have fresh IDs and otherwise matching fields. The raw row count must increase and the selected multiset must remain unchanged. All housing statistics must match exactly. This checks whether counts, lookbacks or regional calculations incorrectly include observations outside the requested geography.

These relations use exact comparisons for counts, month labels, ranks and invariant serialized values. Replication deliberately changes count levels. Its distribution-based invariance applies to medians and the specified relative changes.

Monotonic and Directional Relations

MR05 Broader Price Band

Family: monotonic. An inclusive price band is widened by lowering its minimum by $100,000, bounded at zero, and raising its maximum by $300,000. Other criteria remain fixed. Independent checks must find at least one added observation. Every monthly count and the total must be at least the source value, with a strict increase somewhere. Medians can move either way. This checks price-bound handling and cached selections.

MR06 Restricted ZIP Set

Family: monotonic. A nonempty proper subset of the requested ZIPs is retained. Preflight must find both retained and removed matching observations. Monthly counts and the total cannot increase, and at least one monthly count must fall. When regional comparison is enabled, each retained ZIP keeps its own pooled median. Its rank may change. This checks geography filtering and regional pooling.

MR07 Advanced Window

Family: directional. The twelve-month window is advanced by one calendar month. The dataset must cover both windows and preserve the comparison history of shared months. Every field in the eleven overlapping monthly rows must match exactly, including defined percentage changes. The new total equals the old total minus the outgoing count plus the incoming count. This checks month alignment, window boundaries and lookback handling.

MR08 Price Scaling

Family: directional. Every raw price and any finite price-filter bounds are multiplied by 1.25. Historical comparison months are included. Products must remain exact integer cents and selected membership must stay unchanged. Price levels, price-per-area levels and regional medians scale by 1.25. Counts, percentage changes and ranks remain unchanged. The level tolerance is 0.000003 and the percentage tolerance is 0.000001 percentage points. This checks scaling and consistent filtering.

MR09 Additive Monthly Price Change

Family: monotonic. For a query with no price bounds, $1,000 is added to every observation in one nonempty visible month. Membership must remain unchanged. Its median price increases by $1,000 within $0.000001. Its median price per square foot increases by at least $1,000 divided by the largest selected floor area, allowing 0.000003 for rounding. Counts and other months' price levels stay unchanged. Percentage changes and window summaries can change. This checks month mutation and median calculations.

Unavailable values retain their meaning in every comparison. A test is rejected if its only matches are unavailable values or if the transformation fails its eligibility conditions. A valid test must perform numerical checks.

Experiment Setup

The headline cohort contains thirty distinct eligible base requests for each of the ten relations. Ten criteria groups crossed with three end months produce these requests. The groups include different ZIP selections, property types, bedroom filters and floor-area bands. They cover dense and sparse selections and both odd and even monthly sample sizes.

Every base uses the fixed seeds 104729, 130363 and 155921. For each seed, the source and transformed executions use the same model profile. This produces ninety logical comparisons per relation and 900 comparisons overall. Generator randomness, transformation randomness and model seeds have separate recorded roles.

Eligibility Checks

Before observing model outcomes, an independent raw-CSV preflight checks data validity, month coverage, transformation determinism and membership. It verifies a witness for each relation. Examples include an added observation after price widening, a removed observation after ZIP restriction and a nonidentical row order after shuffling.

A preliminary sparse floor-area band added no observations under price widening. It was rejected before pilot model evaluation and a revised cohort was recorded. Both configurations remain available. The independent checks do not supply the MAS output or replace the paired executions.

Violation Rate and Confidence Interval

A base is complete when all three paired seed outcomes are available. It counts as violating if at least one seed pair violates the required relationship or fails the output contract. The primary rate is the number of such bases divided by the number of completed eligible bases. Each completed row therefore uses a denominator of thirty.

A two-sided 95% Wilson score interval is calculated for that rate. The three fixed seeds represent repeated conditions within a base. They do not create ninety independent base samples. At zero violations in thirty bases, the interval still has an upper bound of about 11.35%. The designed and overlapping slices limit the sampling interpretation of these intervals.

Outcome Classification

A numerical mismatch on an eligible pair is a relation violation. A model parsing, role-policy or required-output failure is an output-contract violation. Invalid test inputs remain outside the rate denominator. Provider unavailability, missing replay evidence and process or tool failures are infrastructure errors and keep a base incomplete until resolved.

The harness can reuse a successful source only when the full dataset, request, prompt, schema, tool, model and seed identity matches. It records that lineage.

Failed sources are not reused. Every transformed follow-up executes afresh. This policy affects how later relations inherit earlier successful source trajectories.

Results

MR	Violating bases	Base rate	95% interval	Violating pairs
MR01	12/30	40.00%	24.59% to 57.68%	20/90
MR02	12/30	40.00%	24.59% to 57.68%	22/90
MR03	16/30	53.33%	36.14% to 69.77%	26/90
MR04	10/30	33.33%	19.23% to 51.22%	14/90
MR05	15/30	50.00%	33.15% to 66.85%	19/90
MR06	12/30	40.00%	24.59% to 57.68%	23/90
MR07	14/30	46.67%	30.23% to 63.86%	24/90
MR08	14/30	46.67%	30.23% to 63.86%	21/90
MR09	14/30	46.67%	30.23% to 63.86%	22/90
MR10	12/30	40.00%	24.59% to 57.68%	19/90

All ten relations completed thirty eligible bases and ninety seed pairs. There were no invalid candidates or unresolved infrastructure errors in the final outcome table. The experiment produced 690 passing numerical pairs and 210 pairs with output-contract violations. Every pair that produced two valid canonical outputs satisfied its declared numerical projection.

The highest base rate was MR03 at 16/30, followed by MR05 at 15/30. MR07, MR08 and MR09 each had 14/30. MR07 was selected as the third relation for diagnosis using the declared identifier tie-break. The confidence intervals overlap substantially.

Execution Counts

The live corpus contains 1,196 terminal analysis attempts and two interrupted captures. The terminal attempts comprise 922 completed analyses and 274 failures. Two failures were retained startup errors during recovery. The final 900 outcomes reference 1,194 distinct terminal analyses, including 272 failed legs. Successful-source reuse accounts for the difference between logical pairs and fresh physical analyses.

Exact failure-signature grouping found 233 scanning repair-guard failures, 37 Reviewer repair-limit failures, two composition failures and the two startup errors. A pair can contain failures on both legs. These physical failure counts must not be added to the primary base numerators.

Terminal leg records contain 10,753 model inputs, 109 refinements and 270 schema-repair requests. A repair request can precede a guard failure, so it does not always represent a completed repair call. Provider telemetry contains 10,763 saved

responses including the interrupted prefixes, one reported transport retry and two inputs whose responses were never captured.

Root Cause Analysis

Shared Failure Sequence

The selected examples for MR03, MR05 and MR07 follow the same coordination failure. The parent queued Price, Supply and Region. It sent the purpose public_comparison to Price, although Price requires purpose statistics. Price copied the invalid purpose and failed validation. The controller then marked Price as awaiting schema repair.

Supply still made its actual model call and returned a schema-valid request with purpose statistics. The controller refused to accept it while the Price repair was pending. The execution stopped with a role-policy error before numerical tool execution or aggregation. Each selected failure contains four captured model decisions.

This sequence identifies a role-specific instruction error followed by a conflict between queued delegation and the repair policy. The three selected relations share this mechanism. Their ranking does not establish three separate bugs.

MR03 Observation Replication

The selected follow-up uses the first base and seed 104729. Independent raw checks confirm 34,560 original rows and 69,120 transformed rows, with fresh record identities and exact twofold multiplicity. Over the selected required months, observations increase from 2,880 to 5,760. The source reuses a successful MR01 execution under the same complete identity.

The follow-up stops at the invalid Price purpose and pending-repair guard. It never produces a final numerical output, so the pair is an output-contract violation. The raw replication witness supports the relation's preconditions. It provides no numerical counterexample to median invariance.

MR05 Broader Price Band

The selected base under seed 104729 fails on its source execution. The same purpose and repair sequence occurs before the transformed run could explain the failure. Independent filtering confirms 2,149 source observations and 2,880 after widening over the required months, including 731 added observations.

The broader selection is a proper superset and meets the count relation's conditions. The observed source failure supports a coordination diagnosis. It cannot establish that widening a price band caused the failure or reduced a count.

Alternative Explanations

Record identity, transformation witnesses, selected membership and the first invalid model response were checked. Those checks support the data and relation conditions for these examples. Exact offline playback reproduces the same failed

control path. The demonstrated cause occurs before any statistical aggregation in the selected runs.

Further Diagnosis

MR07 Advanced Window

The selected follow-up uses the first base and seed 130363. Its source reuses a successful MR01 execution. Independent checks show identical raw observation multisets for the eleven overlapping months. There are 240 outgoing and 240 incoming observations.

The follow-up repeats the invalid Price purpose and pending-repair sequence. The error prevents a final output, so the monthly overlap relation has no complete numerical pair to compare. The valid overlap witness and matching four-decision playback support the same coordination diagnosis as the other selected examples.

Reduced Reproduction

Each representative failure replays from a manifest, terminal record and four original response cassettes. Saved tool results, aggregate outputs and final answers are absent from these reduced inputs. The current Fast Agent workflow still reaches the same failure without network calls.

The reduction removes unnecessary persisted replay dependencies. It retains the complete dataset and exact semantic identities. A smallest failing dataset was not established. Replay did not collect another model sample.

Other Observed Failures

Some Reviewer calls returned the decision schema itself instead of a review decision. The same response format recurred after the allowed repair. Playback reproduced the failure after current tools had executed. These runs ended without an accepted review.

Two composition failures omitted a required caveat after review acceptance. In an inspected example, the selected facts were available and the missing caveat was UNAVAILABLE_LOOKBACK. The parent's shortened hint may have contributed, but the child also received the complete trusted context. One execution cannot isolate the unique cause of that omission.

A separate natural failure occurred after a real refinement. An initial aggregate existed, but a later Price failure stopped the second pass. The system correctly withheld a final answer. Its repair event carried an earlier raw iteration tag. The interface and Langfuse show the later pass using an explicitly explained label derived from the active plan, while retaining the original tag.

Possible Changes

Role-specific purpose hints and completing a pending repair before accepting another queued role are candidate changes. The original corpus remains unchanged. These changes have not been evaluated in a new campaign, so their effect on the observed rates is unknown.

Successful-source reuse and overlapping criteria also affect the rankings. A later relation may inherit an earlier successful source and run only a fresh follow-up. The table describes the declared execution policy on this cohort.

Reproduction and Verification

Live runs capture the model inputs, responses, tool decisions and terminal outcome. Offline reproduction supplies the recorded model decisions through Fast Agent PlaybackLLM and reruns the current controller, statistical tools and MR comparisons. It recomputes the result table and confidence intervals.

Each replay matches the prompt, role schema, tools, dataset, model profile, seed, iteration and preceding receipts. Missing or changed evidence raises a replay error. Network connections are blocked during production playback. Integration tests also replay executions after removing saved numerical results, final answers and source traces.

A clean checkout of committed code passed locked offline setup, deterministic data checks, the Python and frontend checks, framework playback and all 900 comparison replays. The final audit verified 1,194 physical replay analyses and 10,753 recorded decisions. The replay made no fresh model calls and matched the live case semantics.

Replay Corrections

An earlier full replay correctly failed comparison because failed legs lost their seed attribution. Live execution writes cassettes, while playback consumes existing cassettes. A replay-only correction now checks the exact original identity, complete consumed sequence, failure and blocked-network evidence before attributing the historical seed. Each uncorrected record remains saved.

Long replay runs also retained excessive memory. The current delivery driver uses one spawned worker at a time and replaces it after at most sixteen analyses. This resource wrapper calls the same production service. The failed full attempt and a stopped 210-comparison rehearsal remain preserved.

Local Demonstration

Local Docker Langfuse displays the recorded parent and child trajectories. Actual readback verified 1,196 terminal trajectories and 35,356 observations. Browser checks covered normal completion, natural feedback, schema repair and failure. The two interrupted prefixes remain separate from the terminal trajectories.

Desktop and mobile browser checks covered all ten MR summaries, trace inspection, criteria handling and the empty-result state. An actual interface replay recomputed a captured analysis and exported its twelve canonical rows. The final checks passed 106 Python tests, generated-contract checks and frontend tests, build and lint checks.

Command	Purpose
make setup / make data	Install locked dependencies and verify the generated snapshot.
make dev	Start the local application.
make evaluate-live	Run or resume the fixed live cohort, or verify a completed corpus.
make reproduce	Recompute all 900 comparisons offline after setup.
make check	Run Python, frontend and generated-contract checks.

Limitations and AI Use

Limitations

The dataset has regular inventory, designed price patterns and complete clean month coverage. It excludes real listing behavior, addresses and realistic missingness. Passing these relations does not establish the accuracy of predictions about the Dallas housing market.

The experiment uses thirty designed bases per relation, overlapping slices, three fixed seeds and one local model on one machine. Successful-source reuse creates further dependence. The Wilson intervals are nominal summaries for this cohort. They do not establish representative failure probabilities for arbitrary users, models or markets.

Observation counts differ from unique-property counts. Asking prices differ from completed sale prices. The first-to-last statistic compares the endpoints of the visible window, while year-over-year change uses a separate historical month. Those definitions determine which relations are valid.

Shared implementation helpers can contain common faults. Independent raw-data checks, hand-calculated unit cases and mutation controls reduce that risk but cannot prove every output correct. The 690 passing numerical pairs support the tested relationships. Output-contract failures still limit completion of the full agent workflow.

A Mac reboot interrupted the live campaign after 743 resolved comparisons. Saved results were checked intact, but the temporary main console was lost. Recovery restored the pinned Ollama version and added an adapter that normalizes only unchanged parameter ordering after exact value checks. Two startup errors and two partial captures remain saved. A recovery command also ran after a failed preflight and was stopped. Unknown usage for two unanswered inputs remains unknown.

AI Use Reflection

Codex was used to develop the architecture, data generator, contracts, statistical tools, agent integration, test harness, interface and report. The initial handoff was also AI generated. AI could produce those designs and code, propose MR hypotheses and draft plausible explanations without executing the application.

The empirical findings required actual execution. The recorded agent responses, feedback loop, paired outcomes, failures and offline reproduction came from runs of the implemented system. An LLM could calculate a confidence interval from supplied counts. The counts themselves required the completed experiment.

Execution exposed errors in AI-assisted work. An initial sparse input made price widening ineffective, which independent eligibility checks caught before evaluation. Replay exposed reordered tool calls and incorrect reuse of a repeated child name. The all-pass pilot missed failed-leg seed bookkeeping, which the full replay rejected. The recovery mistake described above also required correction.

Failed attempts were retained and corrections were documented in NOTEBOOK.md and AI_USE.md. Injected faults and constructed test decisions remain auxiliary controls. They do not contribute to the natural violation counts. The diagnosis separates demonstrated failure mechanisms from proposed fixes that have not been evaluated.